

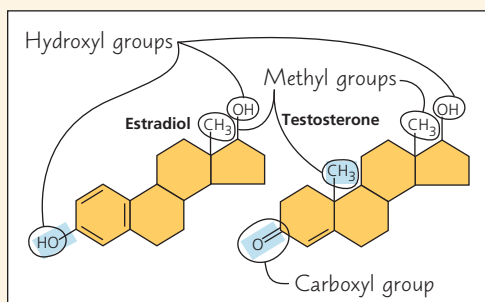
4 Carbon and the Molecular Diversity of Life

KEY CONCEPTS

- 4.1** Organic chemistry is key to the origin of life *p. 57*
- 4.2** Carbon atoms can form diverse molecules by bonding to four other atoms *p. 58*
- 4.3** A few chemical groups are key to molecular function *p. 62*

Study Tip

Label chemical groups: After you have read through Figure 4.9, look through Chapters 4 and 5 for molecules that have the chemical groups shown in that figure. Circle and label the chemical groups you find, as in the following example:



➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 4
- Animation: Diversity of Carbon-Based Molecules
- Animation: Functional Groups

For Instructors to Assign (in Item Library)

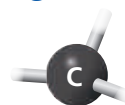
- Activity: Isomers
- Tutorial: Carbon Bonding and Functional Groups



Figure 4.1 The Qinling golden snub-nosed monkeys and other living organisms in this mountainous forest in southwest China are made up of chemicals based mostly on the element carbon. Of all chemical elements, carbon is unparalleled in its ability to form molecules that are large, complex, and varied, making possible the diversity of organisms that have evolved on Earth.

What makes carbon the basis for all biological molecules?

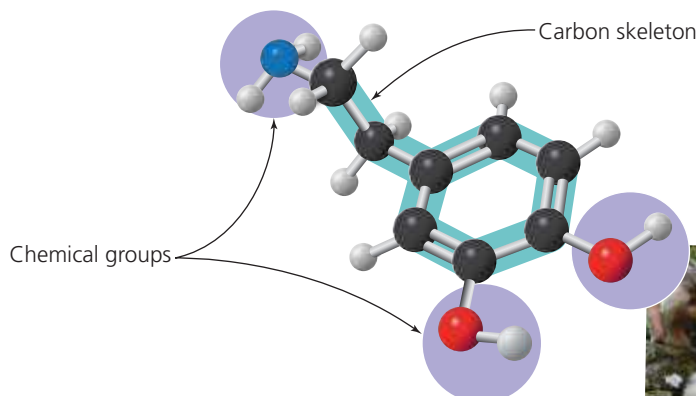
Carbon can form four bonds, and therefore can bond to up to four other atoms or groups of atoms.



Carbon can bond to other carbons, resulting in carbon skeletons. Carbon also commonly bonds to

- H** hydrogen,
- O** oxygen, and
- N** nitrogen.

The properties of a carbon-containing molecule depend on the arrangement of its **carbon skeleton** and on its **chemical groups**.



The signaling molecule shown here, **dopamine**, has many functions, including promoting mother-infant bonding.

